Kevin Benny Mathew

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Education

Expected 2027	B.E. (Hons.) Electronics and Communication Engineering BITS Pilani, Goa Campus	**CGPA: 8.19/10** (through Semester 6)
May 2023	ISC (Class XII) – The Bishop's School, Camp	**97.6%**
May 2021	ICSE (Class X) – St. Mary's School, Pune	**99.0%**

Work Experience

Research Intern – Max Planck Institute for Intelligent Systems Aug 2026 – Present

- Contributing to Dr Rachael Burns' Haptic Empathetic Robot Animal (HERA) project.
- Developing tactile sensing methods to distinguish robot self-touch from external contact, preventing self-touch from interfering with human–robot interaction.
- Updating the tactile sensing system to integrate fully 3D-printed force sensors.

Summer Intern – India Meteorological Department May 2025 – Aug 2025

- Worked on sensor fusion and algorithm development.
- Implemented custom fuzzy logic and Kalman filters for high-accuracy temperature sensing.
- Developed embedded code for ESP8266 hardware using C++ and PlatformIO.
- Conducted data analysis using Python and documented results for research publication.

Projects

Development of Flexible Tactile Sensors Dec 2025 – May 2026

- Replicated existing tactile sensor designs and helped develop a custom flexible piezoresistive force sensor under Dr Tushar Sakorikar.
- Used silicone (PDMS) with carbon nanotubes (CNTs) and reduced graphene oxide (rGO) to sense force through changes in electrical resistance.

Low-Cost Electromechanical Tensile Tester May 2025 – Apr 2026

- Developed a horizontal tensile tester costing under INR 25,000 for soft-material characterisation under Dr Tushar Sakorikar.
- Executed end-to-end development, including mechanical design, 3D modelling, and electronics architecture.
- Integrated motion control using Teensy microcontrollers, stepper motors, and a custom UI.

Project Vulcan: Humanoid Robot Feb 2024 – May 2026

- Led the bionic arm subsystem, developing a 7-DOF biomimetic robotic arm emulating human movement.
- Improved internal mechanical systems for facial movements, specifically eye and mouth movements.
- Implemented amplifiers for microphones located in the robot's ears.

Project Kratos: Mars Rover Aug 2024 – Feb 2026

- Served as a core member of the electronics subsystem; migrated the code stack from ROS Noetic to ROS 2 Humble.
- Redesigned the electronics box for modularity, easier assembly, and improved debugging.

Leadership & Volunteering

Research Head – Electronics and Robotics Club (ERC), BITS Goa Apr 2025 – Apr 2026

- Guided **75+** students across multiple robotics projects (Vulcan, Snakebot, Quadruped, Automated Foosball, Automated Air Hockey).
- Managed project progress and integrated electronics, mechanical, and automation verticals.
- Organised and designed a ROS 2-based hackathon with **200+ participants nationwide**.

Committee Member – Sandbox Innovation Lab, BITS Goa Feb 2025 – May 2026

Maintained 3D printers and laser cutters; managed tech club purchases.

Volunteer – Abhigyaan, BITS Goa Oct 2023 – May 2026

Tutored children of campus non-teaching staff (K–12).

Technical Skills

Programming	C++, C, Python, Java, Verilog
Robotics & Embedded Systems	ROS/ROS 2, Microcontrollers (ESP, Arduino, STM32)
Design & Engineering Tools	KiCad, SOLIDWORKS, Onshape, MATLAB
Development Tools	Git, Linux, PlatformIO, Android Studio
Key Courses	Control Systems, Microprocessors & Interfacing, Digital Signal Processing, Object-Oriented Programming